

ANTIGEN & ANTIBODY

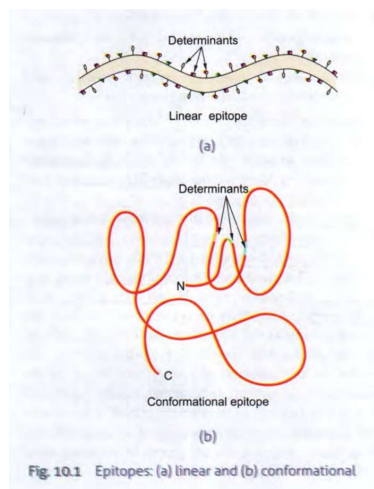
and Antigen–Antibody Reactions

Microbiology (Immunology) • Ananthanarayan & Paniker, Chapters 10, 11 & 12

PART A — ANTIGENS

1. Definitions

- **Antigen:** any substance which, when introduced parenterally, stimulates production of an antibody with which it reacts **specifically** and in an **observable manner**. More precisely, substances recognised by BCRs and by TCRs (along with MHC).
- **Immunogen:** substance that induces a detectable immune response (humoral or cellular). **All immunogens are antigens, but not all antigens are immunogens.**
- Two attributes of antigenicity: **immunogenicity** (induces response) and **immunological reactivity** (reacts specifically with antibody / sensitised cells).
- **Complete antigen:** has both attributes. **Hapten** (Gk *haptēin* = to fasten): cannot induce antibody by itself but reacts specifically with antibody; becomes immunogenic when coupled to a large **carrier** molecule.
- **Complex hapten** = polyvalent, can precipitate. **Simple hapten** = univalent, non-precipitating, but can **inhibit** precipitation.
- **Epitope (antigenic determinant):** smallest unit of antigenicity; ~4–5 amino acids or sugar residues. **Linear/sequential** epitope → recognised by **T cells**; **conformational** epitope → recognised by **B cells**.
- **Paratope:** the combining site on the antibody corresponding to the epitope. Bacteria/viruses carry many epitopes → **antigenic mosaic**.



2. Determinants of Antigenicity

Factor	Key points
Size	Very large molecules (haemocyanin, MW 6.75 million) highly antigenic; MW < 5000 non-antigenic. Small molecules can be made antigenic by adsorbing onto bentonite/kaolin.
Chemical nature	Most antigens are proteins and polysaccharides ; lipids & nucleic acids less antigenic. Structural diversity needed → proteins (20 amino acids) better than polysaccharides. Gelatin is non-immunogenic (structural instability).
Susceptibility to tissue enzymes	Must be metabolised. Polystyrene latex & D-amino acid polypeptides are not antigenic; L-amino acid polypeptides are.
Foreignness	Only "non-self" is antigenic. Ehrlich — horror autotoxicus . More distant species → more antigenic.
Antigenic specificity	Stereochemical basis (Obermayer & Pick, Landsteiner). Even ortho/meta/para position & dextro/levo isomers change specificity.

Types of tissue specificity

- **Species specificity:** used in forensic identification of blood and seminal stains; reflects phylogenetic relationship.
- **Isospecificity (isoantigens):** present in some members of a species, e.g. **human blood group antigens** → transfusion, isoimmunisation in pregnancy.
- **Autospecificity:** self-antigens normally non-antigenic; exceptions = **sequestered antigens** (eye lens protein) and antigens that develop late (**sperm**).
- **Organ specificity:** e.g. brain antigens shared by sheep and man → **neuroparalytic complications** after sheep-brain anti-rabic vaccine.
- **Heterophile (heterogenetic) antigens:** same antigen in unrelated species, e.g. **Forssman antigen** (absent in rabbit). Basis of **Weil-Felix**, **Paul-Bunnell** and **cold agglutinin** tests.

3. T-Dependent vs T-Independent Antigens & Superantigens

Feature	T cell-dependent (TD)	T cell-independent (TI)
Nature	Mostly soluble proteins	Type 1: LPS; Type 2: polysaccharides, polymeric proteins
Antigen processing / MHC	Yes (ternary complex TCR–Ag–MHC)	No (binary complex mIg–Ag)
Isotype switching	Yes	Type 1 no; Type 2 limited
Memory	Yes	No
Antibody produced	Full range: IgM, IgG, IgA, IgE	Limited to IgM and IgG3
Polyclonal activation	No	Type 1: yes (high dose)

- **Superantigens:** proteins (e.g. **staphylococcal enterotoxins**) that bind **outside the antigen-binding groove** — to the lateral aspect of **TCR β chain** and MHC II — activating very large numbers of T cells regardless of specificity → massive **IL-2** & cytokine release.
- Examples: *S. aureus* enterotoxins (food poisoning, **toxic shock syndrome**), Group A streptococcal pyrogenic exotoxins, HIV Nef, rabies nucleocapsid.
- **Innate recognition:** PAMPs recognised by **PRRs** — Toll-like receptors (13 TLRs), scavenger receptors (CD36, CD68, SRB-1) and mannose receptors.

★ GOLDEN ONE-LINERS

- Smallest unit of antigenicity = **epitope**; its counterpart on antibody = **paratope**.
- Hapten = antigenic but **not immunogenic**; needs a carrier.
- Molecules with MW < **5000** are generally non-antigenic; **gelatin** is a classic non-immunogenic protein.
- TI antigens give only **IgM and IgG3**, no memory.
- Superantigen binds **TCR β chain**, outside the peptide groove.

PART B — ANTIBODIES (IMMUNOGLOBULINS)

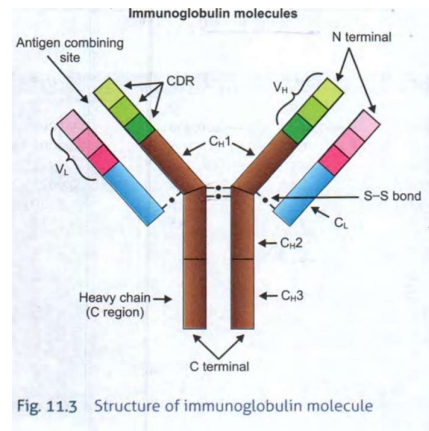
1. Introduction

- Antibodies are **glycoproteins** that bind a specific epitope; present on B-cell membrane and **secreted by plasma cells**.
- **Tiselius (1937)** separated serum proteins by electrophoresis; Tiselius & Kabat (1938) showed antibody activity lies in the **gamma globulin** fraction → "immunoglobulin" (WHO, 1964).
- Immunoglobulins = **20–25% of serum proteins**. **All antibodies are immunoglobulins, but all immunoglobulins may not be antibodies.**

2. Structure

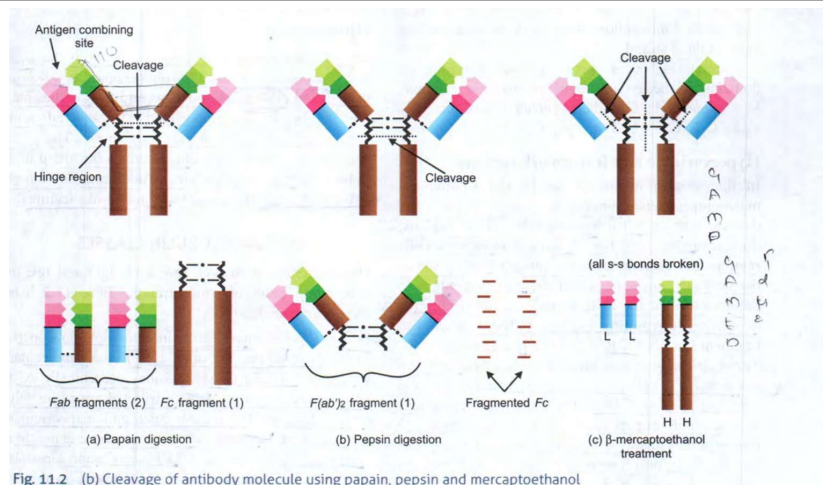
- Four polypeptide chains: **2 identical heavy (H)** and **2 identical light (L)** chains joined by disulphide bonds (Y-shaped).
- **Heavy chains** are class-specific: γ (IgG), α (IgA), μ (IgM), δ (IgD), ϵ (IgE).
- **Light chains** are of two types — **kappa (κ)** and **lambda (λ)** (Korngold & Lapari). One molecule has either κ or λ , **never both**. Human ratio $\kappa:\lambda$ = **60:40**.
- **Variable region** (first ~110 amino acids at N-terminus; VL, VH) → antigen binding. **Constant region** (CL, CH) → effector functions (C-terminal of H chain).

- **Domains:** units of ~110 amino acids, each with an intrachain disulphide loop of ~60 amino acids. L chain = VL + CL; H chain = VH + 3 or 4 CH domains (CH1–CH4). Folded as the **immunoglobulin fold** (β-pleated sheet sandwich).
- **Hypervariable regions / CDRs:** 3 in VH + 3 in VL = **6 CDRs per Fab**; they form the antigen-binding site. Less variable parts = **framework regions**.
- **Hinge region:** between CH1 and CH2, rich in **proline and cysteine**, flexible. Present in γ, δ, α; **absent in μ and ε** (they have an extra CH domain instead).



Enzyme digestion (Porter, Edelman, Nisonoff)

Agent	Products	Key property
Papain (+ cysteine)	2 Fab + 1 Fc (3.5 S)	Fab binds antigen but cannot precipitate (univalent); Fc crystallises in cold
Pepsin	1 F(ab')₂ (5 S) + small Fc fragments	F(ab') ₂ is bivalent → precipitates antigen
β-mercaptoethanol	2 H + 2 L chains	Breaks all disulphide bonds



3. Immunoglobulin Classes

Property	IgG	IgA	IgM	IgD	IgE
Sedimentation (S)	7	7*	19	7	8
Molecular weight	150,000	160,000	900,000	180,000	190,000
Serum conc. (mg/ml)	12	2	1.2	0.03	0.00004
Half-life (days)	23	6	5	2–8	1–5
Classical complement	++	–	+++	–	–
Placental transfer	+	–	–	–	–
Heat stable (56 °C)	+	+	+	+	– (labile)

* IgA may occur in 7 S, 9 S and 11 S forms. Values from Table 11.2 of the textbook.

IgG

- Main serum Ig (~80%); normal level 8–16 mg/ml; half-life **23 days**. Distributed equally intra- and extravascularly.
- **Only Ig that crosses the placenta** → natural passive immunity of newborn.
- "General purpose" antibody: complement fixation, precipitation, neutralisation of toxins & viruses; enhances phagocytosis; late antibody (appears after IgM).
- 4 subclasses — IgG1 (65%), IgG2 (23%), IgG3 (8%), IgG4 (4%). Passive IgG suppresses homologous synthesis → basis of **anti-Rh(D) IgG** to prevent isoimmunisation.

IgA

- Second most abundant (**10–13%**); half-life 6–8 days. Major Ig in **colostrum, saliva, tears**.
- **Serum IgA** = monomer (7 S). **Secretory IgA (SIgA)** = dimer joined by **J chain** + **secretory component** (made by **epithelial cells**, protects from proteolysis); 11 S, MW ~400,000.
- Provides **local (mucosal) immunity**; prevents adherence of microbes. Activates **alternative** complement pathway only. Subclasses IgA1 and IgA2 (IgA2 dominant in secretions).

IgM

- **5–8%** of serum Ig; 19 S, MW 900,000–1,000,000 → "**millionaire molecule**". **Pentamer** with J chain; theoretical valency **10**, effective valency **5**.
- **Phylogenetically oldest, first to be synthesised** by fetus (~20 weeks) and first in primary response. 80% intravascular → protects against **bacteraemia**.
- One IgM molecule lyses as well as **1000 IgG** molecules; 500–1000× better opsonisation, ~20× better agglutination; weaker at neutralisation.
- Destroyed by **0.12 M 2-mercaptoethanol** (differentiates IgM from IgG). Isohaemagglutinins (anti-A, anti-B), typhoid "O" antibodies and reagin (syphilis) are IgM.

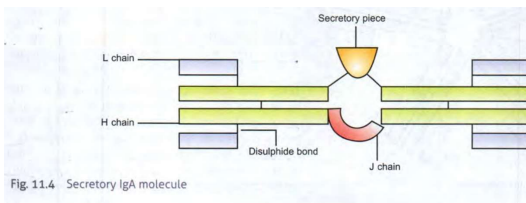


Fig. 11.4 Secretory IgA molecule

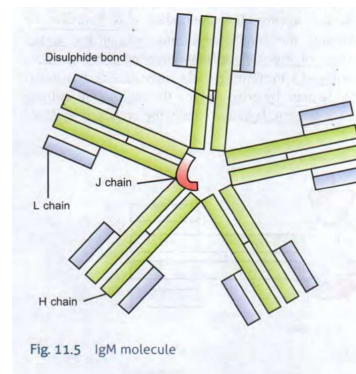


Fig. 11.5 IgM molecule

IgD and IgE

- **IgD**: ~3 mg/100 ml; half-life ~3 days. With monomeric IgM, acts as **antigen receptor on surface of unstimulated B cells**.
- **IgE**: discovered by **Ishizaka (1966)**; 8 S, trace amounts in serum. **Heat labile** (56 °C, 1 h), **homocytotropic** — binds **mast cells**; mediates **Prausnitz–Küstner reaction** and **type I (anaphylactic) hypersensitivity**; raised in atopy and **helminthic infection**. Does not cross placenta or fix complement.



MEMORY AID

- Where each class works: **IgG** — **body fluids**; **IgA** — **body surfaces**; **IgM** — **bloodstream**; **IgE** — **reaginic hypersensitivity**; **IgD** — **B-cell surface receptor**.
- Serum order of concentration: **G > A > M > D > E** — "**GAMDE**".
- Heavy chains without hinge: **M and E** — "**ME** has no hinge, gets an extra domain".

4. Abnormal Immunoglobulins & Ig Specificities

- **Multiple myeloma**: monoclonal plasma-cell proliferation → M protein; urine **Bence Jones proteins** (light chains, either κ or λ only) — **coagulate at 50 °C, redissolve at 70 °C**. **Waldenström's macroglobulinaemia** involves IgM-producing cells.
- **Heavy chain disease**: overproduction of Fc parts of heavy chains (α , γ , μ types). **Cryoglobulinaemia**: gel/precipitate on cooling serum, redissolves on warming (mostly IgM or mixed).

Type	Location / basis	Present in
Isotype	Constant region of heavy chain (classes & subclasses)	All members of a species
Allotype	Allelic differences of 1–4 amino acids	Some members of a species
Idiotype	Variable region / paratope (sum of idiotopes)	Unique to each antibody clone; basis of Jerne's network hypothesis

- **Class switching:** on stimulation the H-chain **VDJ unit** joins a different CH gene; depends on **switch regions, switch recombinase and cytokines (IL-4)**. Defect (**hyper-IgM syndrome**, defective CD40 signalling) → low IgG & IgA, normal/raised IgM.

CLINICAL CORRELATION

- IgM in fetus/newborn = **intrauterine infection** (syphilis, rubella, HIV, toxoplasmosis), since IgM does not cross the placenta.
- IgM in serum indicates **recent infection**; IgG indicates past infection or late stage.
- Anti-Rh(D) IgG given to Rh-negative mother after delivery prevents isoimmunisation.
- IgM deficiency is often associated with **septicaemia**; raised IgE in atopy and worm infestation.

★ GOLDEN ONE-LINERS

- Only Ig to cross placenta — **IgG**. Largest — **IgM**. Longest half-life — **IgG (23 d)**. Heat labile — **IgE**.
- Secretory component is made by **epithelial cells**, J chain by **plasma cells**.
- Papain → **2 Fab + Fc**; Pepsin → **F(ab')₂**.
- Bence Jones protein: coagulates at **50 °C**, redissolves at **70 °C**.

PART C — ANTIGEN–ANTIBODY REACTIONS

1. General Features

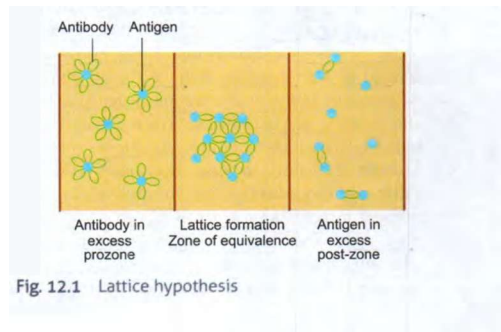
- In-vitro antigen–antibody reactions = **serological reactions**. Uses: basis of humoral immunity & hypersensitivity (in body), laboratory diagnosis, epidemiological surveys.
- **Three stages: Primary** — initial invisible binding by weak forces (van der Waals, ionic, hydrogen bonds), rapid, reversible. **Secondary** — visible events (precipitation, agglutination, lysis, complement fixation, neutralisation). **Tertiary** — in-vivo effects (protection or tissue damage).
- Reaction is **specific** (cross-reactions possible), **whole molecules react, no denaturation**, occurs at the **surface, firm but reversible**, in **varying proportions** (both multivalent).
- **Affinity** = strength of binding between one epitope and one paratope. **Avidity** = overall strength of the multivalent antigen–antibody complex (IgM has low affinity but **high avidity**).
- **Titre** = highest dilution of serum showing an observable reaction. **Sensitivity** = ability to detect minute amounts (few **false negatives**). **Specificity** = reacts only with homologous antigen (few **false positives**). The two are generally in **inverse proportion**.

Reaction	IgG	IgM	IgA
Precipitation	Strong	Weak	Variable
Agglutination	Weak	Strong	Moderate
Complement fixation	Strong	Weak	Negative
Lysis	Weak	Strong	Negative

Table 12.1 of the textbook — comparative efficiency of Ig classes.

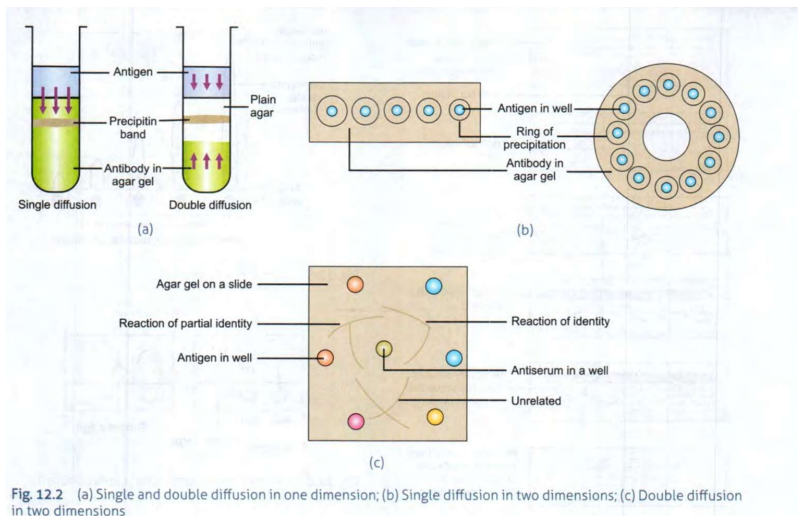
2. Precipitation Reaction

- **Soluble antigen** + antibody + electrolytes (NaCl), suitable temperature & pH → **insoluble precipitate**. If the precipitate stays suspended as floccules → **flocculation**.
- **Zones:** (1) **Prozone** — antibody excess, no visible reaction; (2) **Zone of equivalence** — optimal proportions, maximum precipitation; (3) **Post-zone** — antigen excess. Prozone/post-zone corrected by **serial dilution** of serum.
- **Lattice hypothesis (Marrack, 1934):** multivalent antigen + bivalent antibody form a large lattice only in the zone of equivalence. Also holds for agglutination.



- Sensitive for antigens (as little as **1 µg protein**), less sensitive for antibodies. Applications: forensic identification of blood/seminal stains, food adulterants, **Lancefield grouping** of streptococci, **VDRL**, standardising toxins/toxoids, toxigenicity of diphtheria bacilli.

Type	Principle	Example
Ring test	Antigen layered over antiserum in a narrow tube	Ascoli's thermoprecipitin test; Lancefield grouping
Slide test	Drops mixed on slide → floccules	VDRL (slide flocculation)
Tube test	Flocculation in tubes	Kahn test ; Lf dose of toxin/toxoid
Oudin (single diffusion, 1-D)	Antibody in agar in tube; antigen layered over	No. of bands = no. of antigens
Oakley-Fulthorpe (double diffusion, 1-D)	Plain agar between antibody gel and antigen	—
Radial immunodiffusion (single, 2-D)	Antiserum in agar, antigen in wells → halos; diameter ∝ antigen conc.	Estimation of Ig classes
Ouchterlony (double, 2-D)	Antiserum in centre well, antigens around	Lines fuse = identity; cross = non-identity; spur = partial identity. Elek test



Immunoelectrophoresis & electroimmunodiffusion

- **Immunoelectrophoresis (Grabar & Williams)**: serum proteins separated by electrophoresis, then immunodiffusion against antiserum in a trough → **>30 proteins** identified; used for normal & abnormal proteins in serum and urine.
- **Counterimmunoelectrophoresis (CIE)**: antigen and antibody driven towards each other in opposite directions; result in **30 min, 10× more sensitive** than double diffusion. Detects α-fetoprotein, cryptococcal & meningococcal antigens in CSF.
- **Rocket electrophoresis** (one-dimensional single electroimmunodiffusion): **quantitative estimation of antigens**; precipitin pattern resembles rockets. **Laurell's two-dimensional electrophoresis** quantitates several antigens in a mixture.

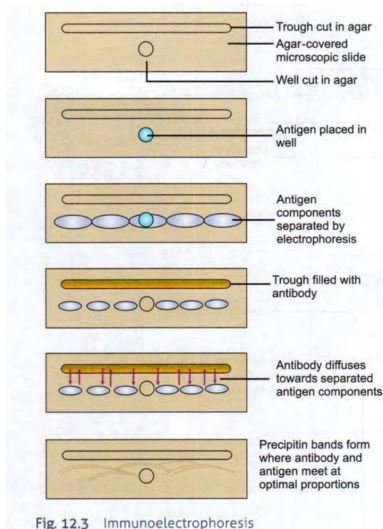


Fig. 12.3 Immunoelectrophoresis

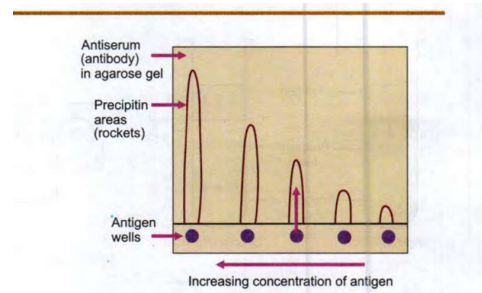


Fig. 12.5 Rocket electrophoresis

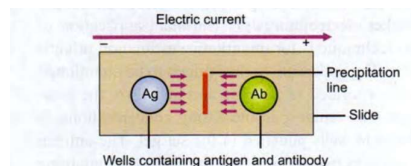


Fig. 12.4 Counterimmunoelectrophoresis

3. Agglutination Reaction

- **Particulate antigen** + antibody + electrolytes → clumping. **More sensitive than precipitation** for detecting antibodies. "Incomplete" (monovalent) antibodies bind but do not agglutinate — act as **blocking antibodies**.
- **Slide agglutination**: rapid; identification of bacterial isolates, **blood grouping & cross-matching**. Always include a **saline control** (to exclude autoagglutination); disregard clumping after 1 minute.
- **Tube agglutination**: quantitative (titre). Used in **Widal test** (typhoid), brucellosis, typhus. **H (flagellar)** antigen → large, loose, fluffy "cotton-wool" clumps; **O (somatic)** antigen → tight, compact "chalk-powder" deposits. Brucella test may show **prozone**.
- **Heterophile agglutination**: **Weil-Felix** (typhus — rickettsiae share antigen with *Proteus*), **Streptococcus MG** test and **cold agglutinin** test (primary atypical pneumonia — human O RBCs at 4 °C, reversible at 37 °C), **Paul-Bunnell** (infectious mononucleosis — sheep RBC agglutinins).
- **Antiglobulin (Coombs) test** (Coombs, Mourant & Race, 1945): detects **incomplete anti-Rh antibodies**. RBCs coated with antibody are agglutinated by **anti-human globulin (Coombs serum)**. **Direct** — sensitisation in vivo (haemolytic disease of newborn); **Indirect** — sensitisation in vitro. Also used in brucellosis.

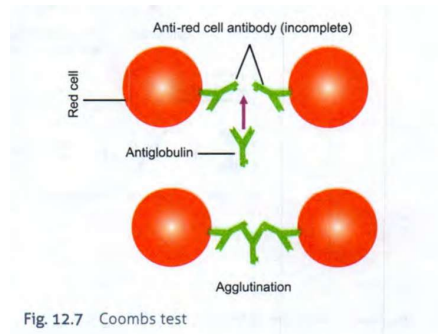


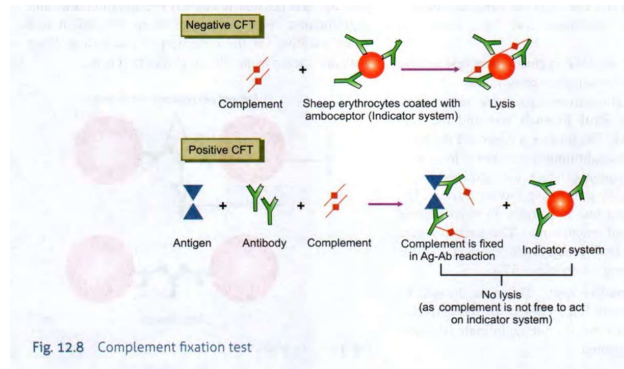
Fig. 12.7 Coombs test

- **Passive agglutination**: soluble antigen adsorbed onto carrier particles (RBC, **latex**, bentonite) → converts precipitation into more sensitive agglutination.
- **Rose-Waaler test** (passive haemagglutination) detects **RA factor** in rheumatoid arthritis. **Latex agglutination** (0.8–1.0 µm polystyrene) for **ASO, CRP, RA factor, hCG**.

- **Reversed passive agglutination:** antibody adsorbed on carrier to detect antigen. **Co-agglutination:** antibody-coated protein A-bearing *S. aureus* as carrier.

4. Complement Fixation Test (CFT)

- **Principle:** antigen–antibody complexes "fix" (consume) complement. Highly sensitive — detects **0.04 mg antibody nitrogen** and **0.1 mg antigen**.
- **Two steps, five reagents:** antigen, antibody (patient serum), **complement (guinea pig serum)**, **sheep RBCs**, **amboceptor** (rabbit anti-sheep RBC antibody). RBC + amboceptor = **indicator (haemolytic) system**.
- Patient serum **inactivated at 56 °C for 30 min** (destroys own complement and anti-complementary factors).
- **1 MHD** (minimum haemolytic dose) of complement = highest dilution of guinea pig serum that lyses one unit volume of sensitised sheep RBCs in a fixed time at 37 °C.
- **Interpretation:** **No haemolysis = positive** (complement fixed); **haemolysis = negative** (complement free). Classic example — **Wassermann reaction** for syphilis.



- **Indirect CFT** — for sera that do not fix guinea pig complement (duck, turkey, parrot, horse, cat); here **haemolysis = positive**. **Conglutinating complement absorption test** uses horse complement and **conglutinin** (bovine serum).
- Other complement-dependent tests: **immune adherence**, **Treponema pallidum immobilisation (TPI)** test (former gold standard for syphilis), **vibriocidal antibody test** (cholera).

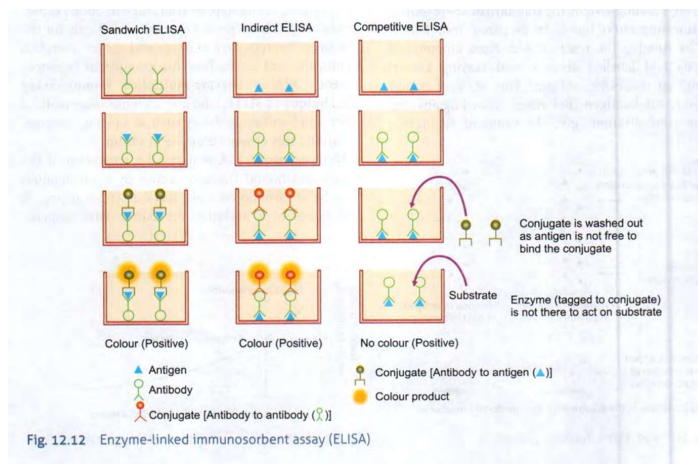
5. Neutralisation Tests & Opsonisation

- **Virus neutralisation:** e.g. **plaque inhibition test** for bacteriophages. **Toxin neutralisation:** exotoxins induce neutralising antitoxins; **endotoxins are not neutralised** by antisera.
- **Schick test** — intradermal diphtheria toxin; tests immunity. **ASO test** — antibodies neutralise haemolytic activity of streptolysin O. **Nagler's reaction** — antitoxin inhibits lecithinase opacity of *C. perfringens* alpha toxin on egg-yolk agar.
- **Opsonin** (Wright, 1903): heat-labile factor (complement) + heat-stable **bacteriotropin** (antibody) that promote phagocytosis. **Opsonic index** = phagocytic activity of patient's blood ÷ that of normal blood.

6. Tests Using Labelled Reactants

- **Radioimmunoassay (RIA):** Berson & Yalow (1959); Nobel Prize 1977. **Competitive binding assay** — labelled and unlabelled (test) antigen compete for limited antibody; B:T ratio read from a standard curve. Detects **picogram (10^{-12} g)** quantities — hormones, drugs, tumour markers, IgE.
- **Enzyme immunoassay (EIA):** most widely used in clinical serology (sensitive, economical, **no radiation hazard**, automation). **Homogeneous EIA** — no separation step, only for haptens/drugs (e.g. **EMIT**). **Heterogeneous EIA** — needs separation & washing; main type is **ELISA**.

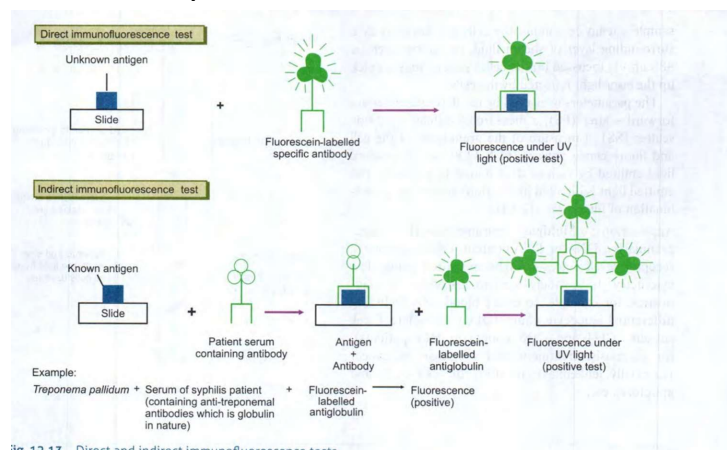
ELISA type	Detects	Principle / example
Direct/Sandwich	Antigen	Antigen sandwiched between capture & enzyme-labelled detection antibodies (rotavirus in faeces; alkaline phosphatase + p-nitrophenyl phosphate → yellow)
Indirect	Antibody	Antigen on well → patient serum → enzyme-labelled anti-human Ig (anti-HIV; HRP + OPD)
Competitive	Antigen / hapten	Sample and standard antigen compete for antibody → inverse curve (more antigen = less colour)
Cassette / Dot blot	Antibody	Rapid (10–15 min), read visually; HIV-1 & 2 antibodies



- **Uses of ELISA:** HIV, hepatitis, EBV, CMV, dengue, TORCH panel, rotavirus, *E. coli* enterotoxin, syphilis, *H. pylori*, food toxins, allergen-specific IgE.
- **Chemiluminescence immunoassay (CLIA):** label = luminol or acridinium esters; light signal measured; fully automated.
- **Western blot (immunoelectroblot):** SDS-PAGE separation → blotting onto **nitrocellulose** → EIA. **Confirmatory test for HIV.**
- **Immunochromatographic test (ICT):** one-step card with **colloidal gold** conjugate; test line + mandatory **control line** (e.g. HBsAg).
- **Immunoelectron microscopy** (hepatitis A, diarrhoeal viruses), **immunoferritin test** (ferritin from horse spleen), **immunoenzyme test** (peroxidase).

Immunofluorescence (Coons, 1942)

- Fluorescent dyes absorb UV and emit visible light. **FITC** — blue-green (apple green); **rhodamine** — orange-red.
- **Direct IF:** labelled specific antibody applied to antigen; e.g. **rabies antigen in brain smears**. Disadvantage: separate conjugate needed for each antigen.
- **Indirect IF:** unlabelled antibody + **fluorescent anti-human globulin**; e.g. **FTA test for syphilis**. One conjugate serves for all antigens. Main drawback of IF: **non-specific fluorescence**.



- **Flow cytometry:** cells in single file by **hydrodynamic focusing**, argon laser (488 nm). **Forward scatter** = cell size, **side scatter** = **granularity**, fluorescence channels for markers. Uses: **CD4/CD8 counts in HIV**, leukaemia typing, DLC, cell cycle & apoptosis.

No textbook diagram of the flow cytometer has been included here to keep the notes short; see Fig. 12.14 in the book.

7. Quick Revision — Test and Its Use

Test	Type of reaction	Used for
VDRL / Kahn	Flocculation (precipitation)	Syphilis
Elek test	Double diffusion (gel precipitation)	Toxigenicity of <i>C. diphtheriae</i>
Widal	Tube agglutination	Typhoid

Test	Type of reaction	Used for
Weil-Felix	Heterophile agglutination	Typhus (rickettsiae)
Paul-Bunnell	Heterophile agglutination	Infectious mononucleosis
Cold agglutinin	Agglutination at 4 °C	Mycoplasma pneumonia
Coombs	Antiglobulin agglutination	Rh incompatibility, incomplete antibodies
Rose-Waaler / Latex	Passive agglutination	RA factor, ASO, CRP, hCG
Wassermann	Complement fixation	Syphilis
Schick / ASO / Nagler	Neutralisation	Diphtheria immunity / streptococcal infection / <i>C. perfringens</i>
Western blot	Immunoelectroblot	Confirmation of HIV
FTA / direct IF	Immunofluorescence	Syphilis / rabies

EXAM FOCUS

- **Zones of precipitation** and **lattice hypothesis** with diagram — frequently asked.
- **Ouchterlony**: fusion = identity, crossing = non-identity, spur = partial identity.
- CFT: know all **five reagents** and that **no haemolysis = positive**.
- ELISA types (sandwich, indirect, competitive) — draw Fig. 12.12.
- Coombs test: direct vs indirect, with an example of each.

MEMORY AID

- Precipitation = **soluble** antigen; Agglutination = **particulate** antigen.
- Prozone = **P**lenty of antibody; Post-zone = plenty of antigen.
- Widal: **H** = **H**airy fluffy clumps; **O** = compact chalky granules.

EXAM QUESTIONS (from the textbook)

Essays

1. Define antigen. Describe the various determinants of antigenicity.
2. Define antibody. Draw a labelled diagram of an immunoglobulin with its different domains.
3. Describe the various classes of immunoglobulins, their properties and functions.
4. Enumerate the types of antigen–antibody reactions; describe the principle and applications of precipitation reactions.
5. Describe the agglutination test and its use in laboratory diagnosis.

Short answers / Short notes

1. TD vs TI antigens; haptens; superantigens; epitopes (linear & conformational)
2. Papain and pepsin digestion of Ig; structure of IgG, IgM and IgA (with diagram); abnormal immunoglobulins; isotypes
3. Complement fixation test; neutralisation test; immunofluorescence; principle & applications of ELISA; sandwich ELISA
4. Immunoelectrophoresis; rocket electrophoresis; Ouchterlony technique; Coombs test; Western blot; flow cytometry; RIA vs ELISA